

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 4: Formulate data-driven web solutions using XML, XSLT, and JSP within the MVC framework. (L4)

Assessment Tool: Class Test

Weightage: 10%

QUESTIONS:

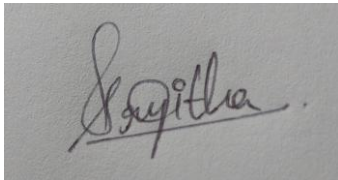
Total Marks: 20

1. Explain the fundamental concepts used in web development by defining the following terms with suitable explanations: **XML, JSP, MVC architecture, XSLT, XML Schema, JSTL, PHP, Database Connectivity, and Parser**. Your answer should clearly describe the purpose and role of each concept in web applications.
2. Explain how the above technologies are interconnected in building a web application. Describe how XML is structured and validated, how JSP and PHP are used for server-side processing, how MVC architecture organizes the application, and how database connectivity and parsers contribute to data processing and communication.

Code of Conduct:

I S.SRIJITHA(192471024)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

1) Fundamental concepts used in web development.

→ 1. XML - Extensible Markup Language.

To store organize and exchange data in a structured format - it uses user-defined tags.

Example:

```
< student >  
  < name > Srijitha < / name >  
  < regno > 192471024 < / regno >  
< / students
```

2. JSP - Java Server Pages

Java used to create dynamic web pages.

It combines HTML with Java JSP elements.

Purpose: JSP receives requests, processes data on the server, and generates dynamic HTML web responses.

3. MVC Architecture - Model-view controller.

model: Handles data and business logic

view: Displays information to the user

controller: receives user requests and connects the model and view.

Purpose: MVC makes application easier to develop maintain, test and modify.

Flow:-

user → controller → model → Database → model →
view → user.

4. XSLT - Extensible stylesheet language transformation.

XSLT is used to transform XML data into other formats, mainly HTML, XML or text.

Purpose:- It separates data from presentation and converts XML into a format that can be easily displayed in a browser.

5. XML Schema (XSD)

- * elements and attributes
- * Data type.
- * Required and optional elements
- * order of elements.

6. JSTL - JSP Standard Tag Library

It provides tags for loops, conditions, formatting, XML processing and other common operations.

Purpose:- JSTL reduces the need to write used to create dynamic website and web application.

7. PHP - Hypertext Preprocessor.

PHP is a server-side scripting language used to create dynamic websites and web application.

- * process form data
- * communicate with databases
- * manage sessions
- * generate dynamic web pages.

8. Database connectivity.

Process of connecting a web application to a database to store, retrieve, update and delete information.

For Java application JDBC [Java Data connectivity] is commonly used.

Purpose:- It allows applications to work with data such as students details, Login information, Products and Marks.

9. Parser:-

A parser is a software component that reads and analyzes structured data according to predefined rules.

For example:-

XML Parser reads and XML documents and identifies its elements, attributes and values.

Purpose:- Parser helps application understand and process data such as XML documents.

2. Interconnection of These Technologies in a web applications.

To create a data-driven web application,

1. XML - for Data Storage and Exchange.

<Student>

<name> Srijitha </name>

<regno> 192471024 </regno>

</Student>

2. XML Schema for validation:-

XML contains correct elements, data type and structure.

XML Document



XML Schema



Valid XML

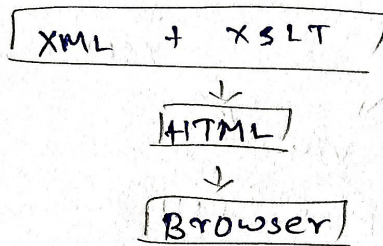
3. Parser for Processing XML.

It contains into a form that the application can process

XML $\rightarrow$ parser $\rightarrow$ Application Data.

4. XSLT for transformation.

Transform it into HTML for display.



5. JSP and JSTL for a Dynamic Pages.

JSP $\rightarrow$ create dynamic web Pages on the server.

JSTL $\rightarrow$ provides standard tags for operations such as conditions and loops.

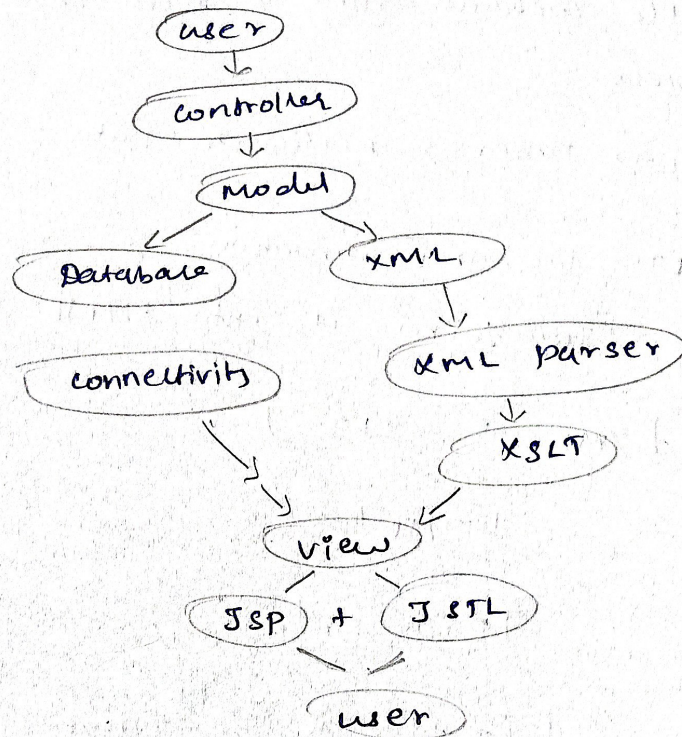
For example: JSP can receive student data from the model and display it as an HTML Pages.

6. PHP for Server-side Processing

It receives user requests, process data, communicates with database, and generates responses.

JSP and PHP are different server-side technologies. Stack rather than combining both for the same server-side layer.

7. MVC organizes the Application.



8. Database connectivity.

Through data connectivity such as JDBC

for Eg:-

